SUMMARY

- The Laplace transformation converts waveforms in the time domain to transforms in the s domain. The inverse transformation converts transforms into causal waveforms. A transform pair is unique if and only if $f(t)$ is causal.
- The Laplace transforms of basic signals like the step function, exponential, and sinusoid are easily derived from the integral definition. Other transform pairs can be derived using basic signal transforms and the uniqueness, linearity, time integration, time differentiation, and translation properties of the Laplace transformation.
- Proper rational functions with simple poles can be expanded by partial fractions to obtain inverse Laplace transforms. Simple real poles lead to exponential waveforms and simple complex poles to damped sinusoids. Partial-fraction expansions of improper rational functions and functions with multiple poles require special treatment. MATLAB software can be used to find Laplace transforms from $f(t)$ and waveforms from $F(s)$. (see Web Appendix C)
- Using Laplace transforms to find the response of a linear circuit involves transforming the circuit differential equation into the s domain, algebraically solving for the response transform, and performing the inverse transformation to obtain the response waveform.
- The initial and final value properties determine the initial and final values of a waveform $f(t)$ from the value of $sF(s)$ as $s \to \infty$ and $s \to 0$, respectively. The initial value property applies if $F(s)$ is a proper rational function. The final value property applies if all of the poles of $sF(s)$ are in the left half plane.

PROBLEMS

OBJECTIVE 9–1 LAPLACE TRANSFORM (SECTS. 9–1, 9–2, 9–3)

Find the Laplace transform of a given signal waveform using transform properties and pairs, using the integral definition of the Laplace transformation, or software applications. Locate the poles and zeros of the transform and construct a pole-zero diagram.

See Examples 9–1, 9–2, 9–3, 9–4, 9–5, 9–6, 9–7, 9–8, 9–9, 9–10 and Exercises 9–1, 9–2, 9–3, 9–4, 9–5, 9–6, 9–7, 9–8, 9–9, 9–10, 9–11, 9–12, 9–13, 9–14, 9–15.

9–1 Find the Laplace transform of $f(t) = 500[1 - e^{-100t}]\, u(t)$. Locate the poles and zeros of $F(s)$.

9–2 Find the Laplace transform of $f(t) = 20 \sin(377t)\, u(t)$. Locate the poles and zeros of $F(s)$.

9–3 Find the Laplace transform of $f(t) = -10\,\delta(t) + 10\,u(t)$. Locate the poles and zeros of $F(s)$.

9–4 Find the Laplace transform of $f(t) = 20[e^{-200t} - 2e^{-100t}]\, u(t)$. Locate the poles and zeros of $F(s)$.

9–5 Find the Laplace transform of $f(t) = A[(B + \alpha t)\, e^{-\alpha t}]\, u(t)$. Locate the poles and zeros of $F(s)$.

9–6 Find the Laplace transform of $f(t) = 0.005[10 - 10\cos(1000t)]\, u(t)$. Locate the poles and zeros of $F(s)$.

9–7 Find the Laplace transform of $f(t) = 5[4\cos(50t) - 5\sin(50t)]\, u(t)$. Locate the poles and zeros of $F(s)$.

9–8 Find the Laplace transform *of* $f(t) = \delta(t) - 200e^{-20t}\cos(200t)\, u(t)$. Locate the poles and zeros of $F(s)$.

9–9 Find the Laplace transform of $f(t) = 10[3 - 5t - 2e^{-15t}]\, u(t)$. Locate the poles and zeros of $F(s)$.

9–10 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams:
(a) $f_1(t) = \left[25e^{-15t} - 20e^{-10t}\right]u(t)$
(b) $f_2(t) = 10[\cos 10t + \cos 20t]u(t)$

9–11 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams.
(a) $f_1(t) = 2\delta(t) + \left[200e^{-200t} + 400e^{-400t}\right]u(t)$
(b) $f_2(t) = \left[15e^{-200t} + 15\cos 500t\right]u(t)$

9–12 9–12 Find the Laplace transforms of the following waveforms. Locate the poles and zeros of $F(s)$. Use MATLAB to verify your results.
(a) $f_1(t) = 5\delta(t) + \left(625t\, e^{-25t}\right)u(t)$
(b) $f_2(t) = \left[10 + 5e^{-10t}(\cos 10t + \sin 10t)\right]u(t)$

9–13 9–13 Find the Laplace transforms of the following waveforms. Use MATLAB to verify your results.
(a) $f_1(t) = 5\delta(t - 2)$
(b) $f_2(t) = 10e^{-50(t-1)}u(t - 1)$
(c) $f_3(t) = 20e^{-50(t-10)}u(t - 10)$

9–14 9–14 Use MATLAB to find the Laplace transform of the following waveform

$$f(t) = \left[10 + 2e^{-10t}\right]u(t) + [5\cos 100(t - 0.05)]u(t - 0.05)$$

9–15 Find the Laplace transforms of the following waveforms.
(a) $f_1(t) = \dfrac{d}{dt}(50e^{-1000t}\cos 200\mathrm{k}t)u(t)$
(b) $f_2(t) = \displaystyle\int_0^t 20e^{-10x}dx + 10u(t) + 20\frac{de^{-10t}}{dt}u(t)$

9–16 Consider the waveform in Figure P9–16.
(a) Write an expression for the waveform $f(t)$ using step and ramp functions.

(b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
(c) Verify the Laplace transform found in (b) by applying the definition of the Laplace transformation in Eq. (9–2) to the waveform $f(t)$ found in (a).

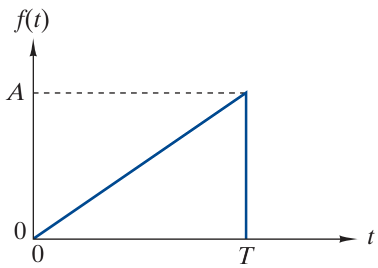

FIGURE P9–16

9–17 Consider the waveform in Figure P9–17.
(a) Write an expression for the waveform $f(t)$ in Figure P9–17 using step functions.
(b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
(c) Verify the Laplace transform found in (b) by applying the definition of the Laplace transformation in Eq. (9–2) to the waveform $f(t)$ found in (a).

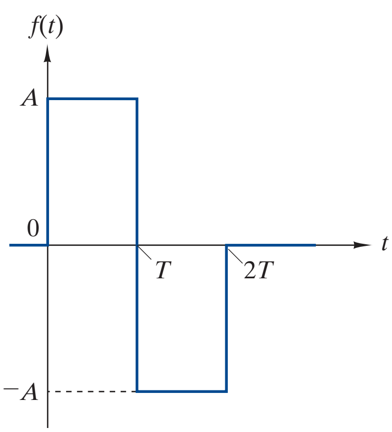

FIGURE P9–17

9–18 For the following waveform: $f(t) = [500 + 100e^{-500t}t \sin 1000t]u(t)$
(a) Find the Laplace transform of the waveform. Locate the poles and zeros of $F(s)$.
(b) Validate your result using MATLAB.

9–19 Consider the waveform in Figure P9–19.
(a) Write an expression for the waveform $f(t)$ in Figure P9–19 using a delayed exponential.
(b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
(c) Verify the Laplace transform found in (b) by applying the definition of the Laplace transformation in Eq. (9–2) to the waveform $f(t)$ found in (a).

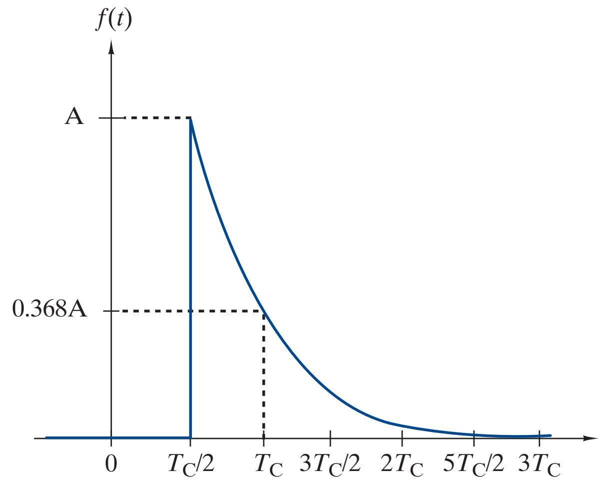

FIGURE P9–19

OBJECTIVE 9–2 INVERSE TRANSFORMS (SECT. 9–4)

(a) Find the inverse transform of a given Laplace transform using partial fraction expansion, basic transform properties and pairs, or using software tools.
(b) Given a pole-zero diagram, find the respective transform.
See Examples 9–11, 9–12, 9–13, 9–14, 9–15 and Exercises 9–16, 9–17, 9–18, 9–19, 9–20, 9–21, 9–22, 9–23, 9–24, 9–25.

9–20 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{50}{s(s+50)}$

(b) $F_2(s) = \dfrac{s+1}{(s+2)(s+3)}$

9–21 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{s+30}{s(s+40)}$

(b) $F_2(s) = \dfrac{(s+10)(s+20)}{s(s+50)(s+100)}$

9–22 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{5000(s+1000)}{(s+500)(s+5000)}$

(b) $F_2(s) = \dfrac{5s^2}{(s+100)(s+500)}$

9–23 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{900}{(s+10)^2+30^2}$

(b) $F_2(s) = \dfrac{3(s+10)}{(s+10)^2+30^2}$

9–24 Find the inverse Laplace transfonns of the following functions and sketch their waveforms for $\beta > 0$:

(a) $F_1(s) = \dfrac{\beta(s+\beta)}{s(s^2+\beta^2)}$

(b) $F_2(s) = \dfrac{s(s+\beta)}{s^2+\beta^2}$

9–25 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{\alpha^2}{s^2(s+\alpha)}$

(b) $F_2(s) = \dfrac{\alpha^2}{s(s+\alpha)^2}$

9–26 Find the inverse Laplace transforms of the following fiinctions:

(a) $F_1(s) = \dfrac{600}{(s+10)(s+20)(s+30)}$

(b) $F_2(s) = \dfrac{2(s+10)}{(s+15)(s+20)}$

9–27 Find the inverse Laplace transforms of the following functions then validate your answers using MATLAB:

(a) $F_1(s) = \dfrac{16s}{(s+3)(s^2+11s+10)}$

(b) $F_2(s) = \dfrac{5(s^2+9)}{s(s^2+25)}$

9–28 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{(s+10000)(s+100000)}{s(s+1000)(s+50000)}$

(b) $F_2(s) = \dfrac{3(s^4+10s^2+4)}{s(s^2+1)(s^2+4)}$

9–29 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{300(s+50)}{s(s^2+40s+300)}$

(b) $F_2(s) = \dfrac{1000s}{(s+5)(s^2+4s+8)}$

9–30 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{16(s^2+256)}{s(s^2+8s+32)}$

(b) $F_2(s) = \dfrac{3(s^2+20s+400)}{s(s^2+50s+400)}$

9–31 9–31 Find the inverse Laplace transforms of the following functions then validate your answers using MATLAB:

(a) $F_1(s) = \dfrac{(s+100)^2}{(s+50)^2(s+200)}$

(b) $F_2(s) = \dfrac{(s+50)^2}{(s+100)^2(s+200)}$

9–32 A certain transform has a simple pole at $s = -20$, a simple zero at $s = -\gamma$, and a scale factor of $K = 1$. Select values for γ so the inverse transform is

(a) $f(t) = \delta(t) - 5e^{-20t}$ **(b)** $f(t) = \delta(t)$ **(c)** $f(t) = \delta(t) + 5e^{-20t}$

9–33 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{s(s+10)(s+20)}{(s+5)(s+30)(s+50)}$

(b) $F_2(s) = \dfrac{(s+1000)(s+5000)}{(s+2000)}$

9–34 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{s^2}{(s+5)}$

(b) $F_2(s) = \dfrac{(s+1000)^2}{(s+2000)^2}$

9–35 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{e^{-5s}(s+20)}{(s+10)(s+30)}$

(b) $F_2(s) = \dfrac{se^{-5s}+20}{(s+10)(s+30)}$

(c) $F_3(s) = \dfrac{s+20e^{-5s}}{(s+10)(s+30)}$

9–36 Use MATLAB to find the inverse transform and plot the poles and zeros of the following function:

$$F(s) = \frac{300s(s^2+30s+400)}{(s+20)(s^3+6s^2+16s+16)}$$

9–37 Use MATLAB to find the inverse transform and plot the poles and zeros of the following function:

$$F(s) = \frac{500(s^3+2s^2+s+2)}{s(s^3+4s^2+4s+16)}$$

9–38 Find the transform $F(s)$ from the pole-zero diagram of Figure P9–38. K is 5.

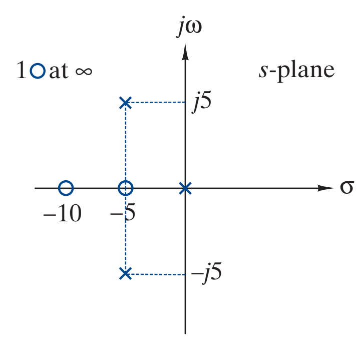

FIGURE P9–38

9–39 Find the transform $F(s)$ from the pole-zero diagram of Figure P9–39. K is 5×10^6.

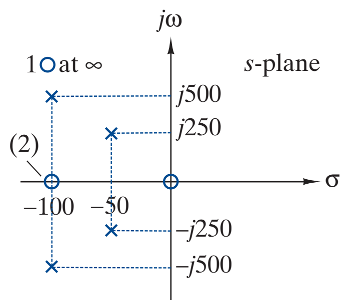

FIGURE P9–39

9–40 Find the transform $F(s)$ from the pole-zero diagram of Figure P9–40. K is 50.

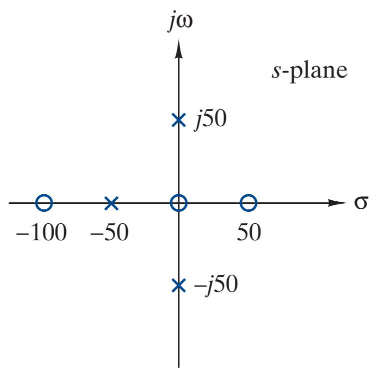

FIGURE P9–40

OBJECTIVE 9–3 CIRCUIT RESPONSE USING LAPLACE TRANSFORMS (SECT. 9–5)

Given a first- or second-order circuit:
(a) Determine the circuit differential equation and the initial conditions (if not given).
(b) Transform the differential equation into the s domain and solve for the response transform.
(c) Use the inverse transformation to find the response waveform.
See Examples 9–16, 9–17, 9–18, 9–19 and Exercises 9–26, 9–27, 9–28, 9–29, 9–30.

9–41 Use the Laplace transformation to find the $v(t)$ that satisfies the following first-order differential equations:

$$\textbf{(a)}\ 250\frac{dv(t)}{dt} + 2500v(t) = 0, \quad v(0-) = 50\ \text{V}$$

$$\textbf{(b)}\ \frac{dv(t)}{dt} + 300v(t) = 600u(t), \quad v(0-) = -150\ \text{V}$$

9–42 Use the Laplace transformation to find the $i(t)$ that satisfies the following first-order differential equation:

$$\frac{di(t)}{dt} + 500\,i(t) = \left[0.100e^{-100t}\right]u(t),\ i(0-) = 0\ \text{A}$$

9–43 The switch in Figure P9–43 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 1\ \text{k}\Omega$, $L = 100$ mH, and $V_A = 15$ V.
(a) Find the differential equation for the inductor current $i_L(t)$ and initial condition $i_L(0)$.
(b) Solve for $i_L(t)$ using the Laplace transformation.

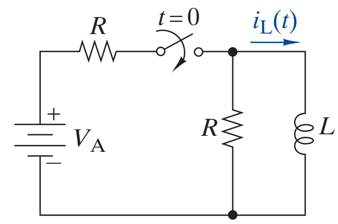

FIGURE P9–43

9–44 The switch in Figure. P9–43 has been closed for a long time and is opened at $t = 0$. The circuit parameters are $R = 50\ \Omega$, $L = 200$ mH, and $V_A = 50$ V.
(a) Find the differential equation for the inductor current $i_L(t)$ and initial condition $i_L(0)$.
(b) Solve for $i_L(t)$ using the Laplace transformation.

9–45 The switch in Figure P9–45 has been open for a long time. At $t = 0$ the switch is closed.
(a) Find the differential equation for the capacitor voltage and initial condition $v_C(0)$.
(b) Find $v_O(t)$ using the Laplace transformation for $v_S(t) = 25\ u(t)$ V.

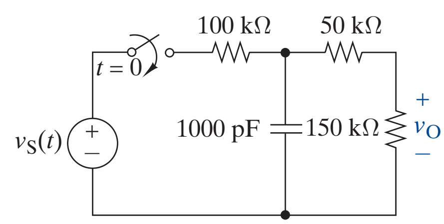

FIGURE P9–45

9–46 Repeat Problem 9–45 for the input waveform $v_S(t) = 169\ [\cos 377t]\ u(t)$ V.

9–47 Repeat Problem 9–45 for the input waveform $v_S(t) = 24\ e^{-1000t}u(t)$ V.

9–48 Use the Laplace transformation to find the $v(t)$ that satisfies the following second-order differential equation:

$$\frac{d^2v(t)}{dt^2} + 20\frac{dv(t)}{dt} + 1000v(t) = 0,\ v(0-) = 20\ \text{V and}\ \frac{dv(0-)}{dt} = 0.$$

9–49 Use the Laplace transformation to find the $v(t)$ that satisfies the following second-order differential equation:

$$\frac{d^2v(t)}{dt^2} + 50\frac{dv(t)}{dt} + 400\ v(t) = 0,\ v(0-) = 0\ \text{and}\ \frac{dv(0-)}{dt} = 1000\ \text{V/s}$$